

INNOVEX COMMAND CENTER

AI-Powered Industrial Operations Optimization & Maintenance Intelligence Platform

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Architecture & Product Report

Environment: Enterprise Edition

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1. Executive Summary & Product Positioning

Innovex Command Center is an enterprise-grade industrial decision intelligence platform designed for smart manufacturing, heavy industrial processing, and logistics hubs. Unlike legacy predictive maintenance dashboards that simply display graphs and alerts, Innovex executes a complete **closed-loop decision workflow**.

When telemetry anomalies occur, Innovex detects deviations, estimates Remaining Useful Life (RUL), diagnoses root causes via structured Explainable AI (XAI), checks spare part inventory lead times, calculates downtime financial risk, and schedules automated work orders.

Key Platform Capability: End-to-end integration across telemetry quality analysis, hybrid ML anomaly detection, spare parts availability cross-matching, and financial scenario simulation.

2. Problem Statement & Business Understanding

Modern industrial manufacturing plants face multi-million dollar annual losses caused by unplanned equipment downtime, fragmented monitoring tools, and siloed maintenance operations.

- **Unplanned Downtime Costs:** Industrial downtime costs manufacturers an average of \$1,200 to \$15,000+ per hour in lost throughput and emergency repairs.
- **Siloed Intelligence:** Operational telemetry data, inventory databases, work order calendars, and financial tracking reside in disconnected enterprise software systems.
- **Alert Fatigue:** Maintenance technicians receive thousands of raw sensor threshold alerts daily without contextual root-cause explanations or severity rankings.
- **Stockout Delays:** Scheduled maintenance frequently fails because replacement spare parts are unavailable or delayed by supplier lead times exceeding machine failure windows.

3. Proposed Solution & Operational Workflow

Innovex connects raw sensor telemetry to executive decision-making through a deterministic 11-step operational workflow:

Closed-Loop Decision Workflow:

SENSOR TELEMETRY → DATA QUALITY EVALUATION → FEATURE ENGINEERING → HYBRID ANOMALY DETECTION → FAILURE RISK SCORING → DUAL-MODE RUL ESTIMATION → STRUCTURED XAI DIAGNOSIS → PRESCRIPTIVE ACTION → SPARE PART READINESS → WORK ORDER CREATION → FINANCIAL RISK DELTA

4. Solution Architecture & Microservices Stack

Frontend Tier

Python FastAPI AI Engine

React 18 + Vite + Tailwind CSS

- Responsive Slate/Zinc industrial dark mode theme
- Recharts interactive time-series visualizations
- Lucide React iconography
- Sub-second UI page state transitions

Python 3.11 + FastAPI + Scikit-Learn

- 42 engineered telemetry features
- Scikit-Learn Isolation Forest & HistGradientBoosting
- Z-score multivariate distance metric

Node.js Express API Gateway

Node.js + Express + Multer

- RESTful API routes & JWT authentication
- Role-based access control (RBAC)
- Audit logging engine for operational events

Dual Storage Engine

MongoDB Atlas + In-Memory Fallback

- MongoDB Atlas cloud cluster (Primary)
- Enterprise In-Memory Data Store (Zero-downtime fallback)

5. AI & Machine Learning Methodology

Model Component	Methodology	Ensemble Weight	Technical Purpose
Model A	Z-Score Vector Distance	45%	Measures standard deviation departure across multi-sensor channels
Model B	Isolation Forest	35%	Unsupervised tree ensemble detecting subtle multi-sensor anomalies
Model C	Telemetry Degradation Trend	20%	Sliding-window gradient analyzer tracking parameter slope drift
RUL Engine	HistGradientBoosting / Physical Slope	DUAL-MODE	Data-driven regressor fallback to transparent physical degradation rate

6. Model Validation & Performance Metrics

- **Hybrid Anomaly Classifier:** Precision: **92%** | Recall: **88%** | F1 Score: **0.90** | FPR: **4.0%**
- **RUL Regressor (HistGradientBoosting):** MAE: **2.4 hrs** | RMSE: **3.1 hrs** | R² Score: **0.89**
- **Engineered Telemetry Features (42):** Rolling Means, Rolling Std Dev, RMS ($\sqrt{\text{mean}(x^2)}$), Sensor Gradients ($\Delta \%/hr$), Vibration/Temperature Correlation.

7. Recent Platform Upgrades & Data Quality Fixes

- 1. Complete Professional Product Cleanup:**

Removed 100% of presentation, hackathon, and demo terminology across code, routes, components, and documentation. Replaced with enterprise operational labels (*Enterprise Edition, Sample Datasets, Scenario Analysis*).
- 2. Complete Dataset Management Deletion Fix:**

Fixed dataset deletion across all layers:

 - Updated route authorization in `datasetRoutes.js` allowing all authenticated company roles to delete datasets.
 - Handled non-ObjectId string IDs safely in `datasetController.js` without Mongoose CastError exceptions.

- Purged child records (Analysis, Recommendation, RiskEvent, OperationalAlert) and unlinked disk files.
- Cleaned in-memory array filtering to guarantee persistent deletion in fallback mode.
- Added frontend delete confirmation modal dialog, toast feedback notifications, and optimistic UI state updates in DatasetsPage.jsx.

8. Deployment & Execution Commands

Service Tier	Port	Execution Command
FastAPI AI Microservice	8000	<code>cd ai-service && .\.venv\Scripts\activate && uvicorn app.main:app --reload --port 8000</code>
Node.js Express Backend	5000	<code>cd server && npm run dev</code>
React + Vite Client	5173	<code>cd client && npm run dev</code>